AI Usage Report
Team Control Number: 2515586
Problem Chosen: 2025 MCM Problem A

1. Specific Applications of AI Tools

1. **ChatGPT-3.5 for Programming and Data Processing Assistance**
    - Used to generate initial code for Python plotting and data distribution functions (e.g., Gaussian distribution). The team debugged and optimized the generated code to ensure it met practical requirements and verified the accuracy of the results.
2. **DeepSeek for Problem Understanding and Literature Retrieval**
    - Assisted in understanding the MCM problem, helping the team quickly grasp the core points.
    - Helped locate key references and explained specialized terms and formulas. For example, the explanation of Archard's wear law:
        - **Reference**:Archard, W. (1953). Research in the Journal of Research in Friction Materials. American Materials Scientists.This paper presents the Archard wear law, which is used in your wear degree model.
          Mathematical Society and Addison-Wesley
          Publishing Company , 1984-1986.
        - **Explanation**: Archard wear law describes the relationship between material wear volume, contact pressure, sliding distance, and material hardness, expressed as $W = K\frac{F_n \cdot d}{H}$, where W is wear volume, K is the wear coefficient, F_n is the normal force, d is the sliding distance, and $H$is the material hardness.
3. **Kimi for Data Compilation and Analysis**
    - Used Kimi to compile and analyze all collected references, data, and team-summarized methodologies, helping the team sort out research ideas.
    - Validated the logic of the team's wear degree model to ensure its rationality and scientific rigor.
4. **DeepSeek for Online Tool Price Comparison**
    - Used DeepSeek to compare prices of relevant tools, helping the team select cost-effective equipment. For example, querying prices of portable 3D scanners and comparing the accuracy and cost of different models.
5. **ChatGPT-4.0 for Language Polishing**
    - Used ChatGPT-4.0 to refine certain sentences, improving the language quality of the paper. For example:
        - Original: "We use some factors to decide how people walk."
        - Polished: "The fuzzy comprehensive evaluation system incorporates four critical factors (U1-U4) to characterize pedestrian flow patterns."

2. Team's Main Responsibilities

1. **Core Model Development**
    - The team independently derived the theoretical framework and formulas for the wear degree model.
2. **Data Collection and Processing**
    - The team collected and processed actual data, verifying and correcting preliminary data generated by AI.
3. **Model Validation and Optimization**
    - The team conducted multiple validations and optimizations to ensure the model's effectiveness and stability in practical applications.
4. **Paper Writing and Chart Creation**
    - The team was responsible for the overall writing of the paper and the creation of charts, ensuring logical clarity, structural coherence, and adherence to academic standards.

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3. Usage Boundaries

1. **Core Work Independently Completed**
    - Theoretical derivation of the wear degree model
    - Determination of the directional index (R_{dir}) threshold
    - Development of on-site measurement plans
2. **AI Output Verification Mechanism**
    - All reference links were manually verified for accessibility.
    - Generated code was tested line by line.
    - Graphical data were cross-verified using Matlab.